

Time-Based Non-Uniform Quantization with Linear-Bias Integrate-and-Fire TEMs

Anshu Arora^{a,1}, Kaluguri Yashaswini^{a,1}, Satish Mulleti^{a,*}

^a*Department of Electrical Engineering, Indian Institute of Technology
Bombay, Powai, Mumbai, 400076, Maharashtra, India*

Abstract

Time encoding machines (TEMs) provide an event-driven alternative to uniform sampling, enabling power-efficient signal acquisition without a global clock. We show that firing intervals in integrate-and-fire TEMs are inherently non-uniformly distributed, rendering uniform quantization (UQ) inefficient. For a class of bandlimited signals, we derive the probability distribution of firing intervals and design a power-law-based non-uniform quantizer (NUQ) tailored to this distribution. Simulation results demonstrate that NUQ significantly outperforms UQ under the same bit budget. We further compare TEMs with non-uniform sampling, where both amplitudes and timings must be quantized, and show that TEM-NUQ achieves lower reconstruction error at half the communication cost. Building on these results, we propose a linear-bias integrate-and-fire TEM that dynamically tracks the input to

*Corresponding author

Email addresses: anshuarora2604@gmail.com (Anshu Arora),
yashkaluguri2005@gmail.com (Kaluguri Yashaswini), mulleti.satish@gmail.com
(Satish Mulleti)

URL: <https://sites.google.com/view/satishmulleti/> (Satish Mulleti)

¹Equal contributoin.

localize firing intervals, enabling more efficient low-bit NUQ. The proposed framework achieves improved rate–distortion performance while maintaining low power consumption and analytical tractability.

Keywords: Integrate-and-fire time encoding machines, asynchronous sampling, low-rate sampling, non-uniform quantization, oversampling control, event-driven sampling, reconstruction error

1. Introduction

Sampling and quantization are fundamental to digital signal processing, enabling continuous-time signals to be represented, stored, and transmitted efficiently. Quantization is typically uniform (UQ) or non-uniform (NUQ): while UQ is simple, it is suboptimal when signal amplitudes are not uniformly distributed, whereas NUQ adapts the resolution to the data statistics and yields improved rate–distortion performance [1].

Conventional architectures rely on uniform sampling, where amplitudes are measured on a global clock and then quantized. This approach is hardware-intensive, power-hungry, and prone to clocking and interference issues [2], motivating event-driven schemes that abandon uniform time grids in favor of signal-dependent sampling. Non-uniform sampling methods such as random sampling [3], level-crossing schemes [4], and time-encoding machines (TEMs) [5–14] encode information in timing rather than amplitudes; in particular, non-uniform sampling (NUS) [3] and TEMs represent signals more efficiently than uniform sampling by exploiting this event-based nature.

17 Among these, the integrate-and-fire TEM (IF-TEM) stands out for its
 18 simplicity and low power consumption [15, 16]. By integrating the input plus
 19 a bias and firing whenever a threshold is reached, IF-TEMs produce time en-
 20 codings whose inter-event intervals adapt to the signal dynamics. They also
 21 allow explicit control of minimum and maximum inter-firing times, enabling
 22 iterative reconstruction algorithms [11], a feature generally unavailable in
 23 classical level-crossing schemes. For perfect reconstruction from TEM en-
 24 codings, inter-firing intervals must remain below the Nyquist interval; this
 25 typically forces oversampling in high-amplitude regions, leading to a wide
 26 range of firing intervals and increased information rate. Since firing inter-
 27 vals must ultimately be quantized, a larger dynamic range directly increases
 28 quantization error for a fixed bit budget.

29 Uniform quantization of IF-TEM intervals has been analyzed in [11],
 30 where the induced error is shown to admit essentially the same upper bound
 31 as amplitude quantization in NUS. In practice, however, firing intervals are
 32 far from uniformly distributed, making UQ fundamentally inefficient and mo-
 33 tivating NUQ schemes that explicitly exploit the statistical structure of time
 34 encodings.

35 In this paper, we study NUQ for IF-TEM and propose a modified IF-
 36 TEM architecture that improves the efficiency of low-bit time encodings. For
 37 a class of stationary, ergodic, bandlimited stochastic processes, we derive the
 38 probability distribution of IF-TEM firing intervals and relate it explicitly to
 39 the amplitude distribution, showing that the induced firing-interval laws are

highly non-uniform and thus naturally suited to NUQ. Using this characterization, we design a power-law-based NUQ that adapts to the firing-interval statistics and achieves lower reconstruction error than UQ under the same bit budget; moreover, when comparing IF-TEM to NUS, where both time and amplitude must be quantized, we show that TEM-NUQ attains superior accuracy at lower communication cost.

Further, we introduce the *Linear-Bias Integrate-and-Fire Time Encoding Machine (LB-IF-TEM)*, in which a piecewise linear bias tracks the input signal². By maintaining a nearly constant integrator input, the LB-IF-TEM reduces the variance of firing intervals while preserving perfect reconstruction guarantees, yielding firing-interval distributions that are particularly well suited to low-bit NUQ. Model-driven quantization approaches for TEMs, with recovery guarantees based on frame theory, have been proposed in [19], achieving up to one-order error reduction; our focus is complementary, as we modify the TEM architecture itself and employ relatively simple NUQ schemes to reduce the information required for signal recovery. Variable-bias IF-TEMs (VB-IF-TEMs) have been used to reduce oversampling and control firing rates [7, 20]; for example, the VB-IF-TEM in [20] limits the oversampling factor to approximately 4.23. While effective in lowering firing rates, such designs impose intrinsic limits on the achievable interval range and restrict flexibility when denser sampling is required. In contrast, the

²Preprints of these works on NUQ and LB-IFTEM together with the supplementary results are available online [17, 18].

61 proposed LB-IF-TEM permits the oversampling factor to be freely selected
62 based on system requirements, while simultaneously concentrating the firing-
63 interval distribution for more efficient quantization.

64 Although linear biasing has been explored previously [21], those schemes
65 rely on signal differentiation and are limited to finite rate-of-innovation mod-
66 els. Here, we extend linear-bias time encoding to general bandlimited signals
67 and couple it with distribution-aware NUQ of the induced time encodings.
68 The resulting framework combines low-power operation, reduced communi-
69 cation overhead, and analytical tractability within a unified time-encoding
70 architecture.

71 The paper is organized as follows. Section 2 introduces notation and
72 the mathematical set-up. Section 3 presents the NUQ design for TEMs
73 and a probability density transformation law relating amplitude and firing
74 distributions; corresponding results are analyzed in Section 4. Section 5
75 introduces and analyzes the LB-IF-TEM, and Section 6 reports experimental
76 results and discusses implications. Section 7 concludes the paper.

77 2. Problem Formulation

78 We model the input as a real-valued, stationary, ergodic, bandlimited
79 stochastic process $\{f(t)\}_{t \in \mathbb{R}} \in \mathcal{B}_{\Omega_0, c}$, with spectral support in $[-\Omega_0, \Omega_0]$ and
80 bounded amplitude $|f(t)| \leq c$. For any $f(t) \in \mathcal{B}_{\Omega_0, c}$, its Fourier transform
81 $F(\Omega) = \int_{-\infty}^{\infty} f(t) e^{j\Omega t} dt$ satisfies $F(\Omega) = 0$ for all $|\Omega| > \Omega_0$. By Shan-
82 non-Nyquist sampling theory, any $f(t) \in \mathcal{B}_{\Omega_0, c}$ can be exactly reconstructed

83 from uniform samples $\{f(nT_s)\}$ if $T_s < T_{\text{Nyq}} = \frac{\pi}{\Omega_0}$, i.e., at a rate $\Omega_0/\pi =$
 84 $1/T_{\text{Nyq}}$. Moreover, signals in this class satisfy $|f'(t)| \leq c\Omega_0 = \epsilon$ [22]. Alterna-
 85 tively, reconstruction from a non-uniform sampling (NUS) set $\{f(t_n), t_n\}_{n \in \mathbb{Z}}$
 86 is possible if $t_{n+1} - t_n < T_{\text{Nyq}}$ [23], but in that case both amplitudes and
 87 sampling locations must be quantized.

88 To analyze NUQ in TEMs, we now consider the IF-TEM as a time-based
 89 encoder for signals in $\mathcal{B}_{\Omega_0, c}$. In a conventional IF-TEM, the input $f(t)$ is
 90 biased to obtain a strictly non-negative signal $f(t) + b$ with $b > c$, which is
 91 then integrated with scaling factor $1/\kappa$. A firing occurs when the integrator
 92 output reaches a fixed threshold Δ , after which the integrator is reset. This
 93 generates a strictly increasing sequence of firing instants $\dots < t_{n-1} < t_n <$
 94 $t_{n+1} < \dots$ related to the signal by

$$\frac{1}{\kappa} \int_{t_{n-1}}^{t_n} (f(t) + b) dt = \Delta. \quad (1)$$

95 The time encodings $\{t_n\}$ thus form a discrete representation of $f(t)$, in con-
 96 trast to the uniform samples $\{f(nT_s)\}$ or the NUS set $\{t_n, f(t_n)\}$. Defining
 97 $T_n = t_{n+1} - t_n$, one obtains

$$T_{\min} = \frac{\kappa\Delta}{b+c} \leq T_n \leq \frac{\kappa\Delta}{b-c} = T_{\max}, \quad (2)$$

98 using (1) and the dynamic range $[-c, c]$ of $f(t)$.

99 Given the intervals $\{T_n\}$, the signal $f(t)$ can be perfectly reconstructed
 100 via an iterative algorithm, provided that $T_n < T_{\text{Nyq}}$ for all n [11]. The

101 parameters κ and Δ must therefore be chosen, subject to (2), to satisfy this
 102 condition; without loss of generality, we set $\kappa = 1$ and treat Δ as a free
 103 design parameter.

104 In practice, the intervals $\{T_n\}$ must be quantized for storage or digital
 105 processing, using either UQ or NUQ [1, 8]. Quantization introduces recon-
 106 struction error, which can be reduced either by increasing the number of bits
 107 or by designing a NUQ that matches the distribution of the time encodings.
 108 Since this distribution depends on both the amplitude law of $f(t)$ and the
 109 IF-TEM parameters, it is essential to relate the amplitude and firing-interval
 110 distributions in order to design an effective NUQ for $\{T_n\}$.

111 **Remark 1** (Deterministic vs. stochastic modeling). *The IF-TEM principle*
 112 *above applies to any bounded, bandlimited signal, and the reconstruction guar-*
 113 *antees hold in a deterministic setting. For quantization analysis, however,*
 114 *we require the statistics of the quantities being quantized. We therefore adopt*
 115 *a stochastic model, treating $\{f(t)\}_{t \in \mathbb{R}}$ as a stationary, ergodic, bandlimited*
 116 *random process with an amplitude distribution. This allows us to define the*
 117 *induced distribution of the firing intervals $\{T_n\}$ and to design a NUQ tailored*
 118 *to these statistics. Ergodicity ensures that ensemble distributions can be ac-*
 119 *curately approximated from single realizations, thereby bridging deterministic*
 120 *and stochastic viewpoints.*

121 Since UQ performance is governed by the interval range $T_{\max} - T_{\min}$, re-
 122 ducing this range improves encoding precision. For NUQ, it is more beneficial
 123 to achieve a concentrated, low-variance firing-interval distribution [24]. This

motivates *range concentration* as a design goal and, later in the paper, a linear-bias strategy $b(t)$ that localizes the variance of firing intervals within a narrower range, thereby improving the quantization efficiency.

3. Distribution of Time Encodings and NUQ Design

In this section, we first derive the distribution of the time encodings and then discuss the NUQ design aspects.

3.1. Theoretical Results

Efficient non-uniform quantization requires knowledge of the underlying statistics of the data being quantized. In our setting, these statistics are determined by the amplitude law of the input process, which we denote by the *ensemble distribution* to distinguish it from empirical histograms obtained from realizations.

For any fixed $t \in \mathbb{R}$, let $\mathbf{F}$ be the random variable corresponding to the amplitude $f(t)$, with ensemble law $P_{\mathbf{F}}(f)$. Equivalently, at any time instant t , the random variable $f(t)$ is distributed according to $P_{\mathbf{F}}$.

The IF-TEM produces a sequence of time encodings $\{T_n\}$ according to (1). We denote by $\mathbf{T}$ the random variable governed by the distribution $P_{\mathbf{T}}(T)$, which is induced from $P_{\mathbf{F}}(f)$ through the IF-TEM mapping. This relationship is formalized below.

Theorem 1 (Ensemble distribution of the time encodings). *Let $\{f(t)\}_{t \in \mathbb{R}}$ be a stationary, ergodic, bandlimited stochastic process with bounded amplitude*

145 $|f(t)| \leq c$ almost surely, and with amplitude distribution $P_{\mathbf{F}}(f)$ supported on
 146 $[-c, c]$. Then the ensemble distribution of the time encodings $\{T_n\}$ generated
 147 by the IF-TEM is

$$P_{\mathbf{T}}(T) = \begin{cases} P_{\mathbf{F}}\left(\frac{\Delta}{T} - b\right) \frac{\Delta}{T^2}, & \frac{\Delta}{b+c} \leq T \leq \frac{\Delta}{b-c}, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

148 where c is the maximum amplitude of the process.

149 *Proof.* The IF-TEM encoding law is $\int_{t_n}^{t_{n+1}} (f(t) + b) dt = \Delta$, which can be
 150 equivalently written as

$$T_n = \frac{\Delta}{\left(\frac{1}{t_{n+1}-t_n} \int_{t_n}^{t_{n+1}} f(t) dt\right) + b}. \quad (4)$$

151 Since $f(t)$ is bandlimited, it is continuous, and by the Mean Value Theorem
 152 for integrals there exists a point $\tau_n \in [t_n, t_{n+1}]$ such that $\frac{1}{t_{n+1}-t_n} \int_{t_n}^{t_{n+1}} f(t) dt =$
 153 $f(\tau_n)$. Substituting into (4) gives

$$T_n = \frac{\Delta}{f(\tau_n) + b}. \quad (5)$$

154 By stationarity, the distribution of $f(\tau_n)$ coincides with the ensemble distri-
 155 bution of $f(t)$ at any fixed time [25, Chapter 9]. Thus, $f(\tau_n) \sim P_{\mathbf{F}}, T_n \sim P_{\mathbf{T}}$.
 156 The probability density function of $\mathbf{T}$ induced by the mapping $g(f) = \Delta/(f +$
 157 $b)$ is obtained via the change-of-variables theorem. Since g is monotone on

158 $[-c, c]$, we have $P_{\mathbf{T}}(T) = P_{\mathbf{F}}(g^{-1}(T)) \left| \frac{d}{dT} g^{-1}(T) \right|$, where $g^{-1}(T) = \frac{\Delta}{T} - b$, and
 159 $\frac{d}{dT} g^{-1}(T) = -\frac{\Delta}{T^2}$. A substitution gives

$$P_{\mathbf{T}}(T) = P_{\mathbf{F}}\left(\frac{\Delta}{T} - b\right) \frac{\Delta}{T^2}. \quad (6)$$

160 Finally, since $\mathbf{F} \in [-c, c]$, the support of T is

$$\frac{\Delta}{b+c} \leq T \leq \frac{\Delta}{b-c}. \quad (7)$$

161 Equations (6) and (7) together establish the claimed distribution. $\square$

162 **Remark 2** (Empirical vs. Ensemble Distribution). *The functional form*
 163 *of $P_{\mathbf{T}}(T)$ provides the ensemble distribution of firing intervals. In practice,*
 164 *however, only a single realization of $\{T_n\}$ is observed.*

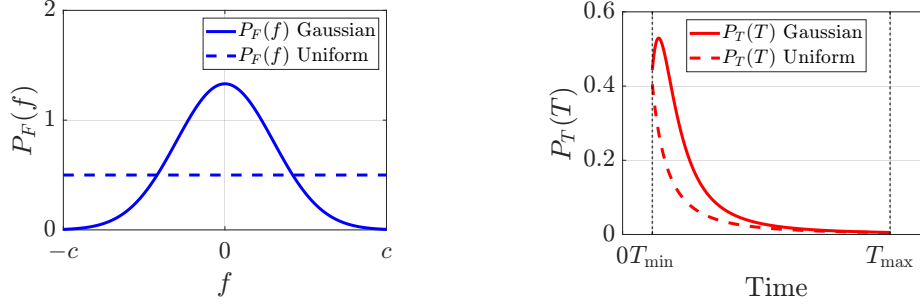
165 *Since $\{f(\tau_n)\}_{n \in \mathbb{N}}$ is stationary and ergodic, and the mapping $g(x) = \frac{\Delta}{b+x}$*
 166 *is measurable and bounded (as ensured by (7)), it follows that the sequence*
 167 *$\{T_n\}$ is also stationary and ergodic*

168 *Thus, by Birkhoff's Ergodic Theorem, for any integrable function φ ,*

$$\frac{1}{N} \sum_{n=1}^N \varphi(T_n) \xrightarrow[N \rightarrow \infty]{a.s.} \mathbb{E}[\varphi(\mathbf{T})]. \quad (8)$$

169 *In particular, for $\varphi = \mathbf{1}_A$ and the Borel set $A \subseteq [\Delta/(b+c), \Delta/(b-c)]$,*

$$\frac{1}{N} \sum_{n=1}^N \mathbf{1}_{\{T_n \in A\}} \xrightarrow[N \rightarrow \infty]{a.s.} P_{\mathbf{T}}(A). \quad (9)$$

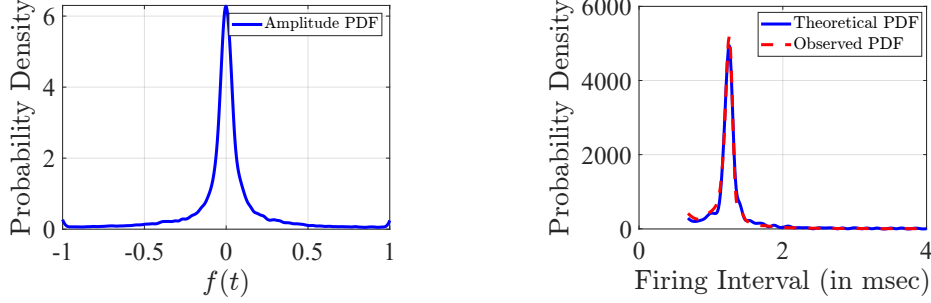


(a): Truncated Gaussian signal distribution, along with uniform amplitude distribution. (b): For corresponding, Gaussian and uniform inter-event distributions, time PDFs $P_T(T)$.

Figure 1: Relationship between the signal distribution and the induced firing time distribution in the IF-TEM framework.

Therefore, the empirical histogram of $\{T_n\}$ converges almost surely to the theoretical law P_T . This ensures that a non-uniform quantizer designed from the ensemble law is valid in practice based on long single realizations.

In Fig. 1, we compare the distributions of signal amplitudes and the corresponding firing intervals obtained via (3) for Gaussian and uniform amplitude laws. A key observation is that the nonlinear mapping of amplitudes to firing intervals inherently distorts the distribution: even when the amplitudes are uniformly distributed, the resulting time encodings are far from uniform. For Gaussian amplitudes, the effect is more pronounced, with firing intervals clustering in specific regions rather than spreading evenly. These concentration patterns make uniform quantization fundamentally inefficient, as it fails to match the true statistical structure of the data. This motivates the use of non-uniform quantization, which allocates finer resolution where encodings are more likely to occur, thereby reducing distortion and improving reconstruction performance.



(a): Amplitude distribution from the simulated signals.

(b): Observed and theoretical distributions of firing times.

Figure 2: A comparison of the empirically observed distribution and the theoretical one. (3) to the PDF in (a).

185 3.2. Practical NUQ Design for IF-TEM

186 To design a NUQ, we need exact knowledge of the distributions of ampli-
 187 tudes. In practice, only the empirical distributions can be determined. Be-
 188 fore discussing NUQ choices, we first assess the closeness of the empirical
 189 distributions to the true ones.

190 To this end, in Fig. 2, we illustrate the distributional amplitude and time
 191 encodings of the IF-TEM outputs. To generate these results, 100 independent
 192 realizations of a bandlimited stochastic process were synthesized as sums of
 193 sinc functions with spectral support up to 50 Hz. Each signal was sampled at
 194 a resolution of $1 \mu\text{s}$ to obtain high-fidelity approximations of the underlying
 195 distributions, where the continuous-time, bandlimited nature of the process
 196 ensures accurate representation of the theoretical models.

197 Fig. 2(a) shows the averaged histogram of the signal amplitudes com-
 198 puted over the 100 independent realizations (signals). This averaging proce-
 199 dure provides an empirical estimate of the ensemble law $P_{\mathbf{F}}(f)$ [25, chapter

7] of the bandlimited stochastic process. Fig. 2(b) presents an overlay of the theoretically derived PDF of the interspike intervals $P_{\mathbf{T}}(T)$, obtained from (3), and the observed distribution of $\{T_n\}$ estimated from the simulated signals as for the amplitudes. The close agreement between the theoretical and simulated distributions validates the analytical derivation of the interspike interval ensemble distribution and confirms the accuracy of the IF-TEM simulations.

Motivated by this characterization, we now design non-uniform quantizers for the firing intervals. In principle, NUQ can be implemented directly using optimization-based methods such as the Lloyd–Max algorithm, or indirectly by applying a suitable transformation followed by uniform quantization. The latter approach is particularly well-suited for the present application, since power-law-based companding schemes are inherently more robust to distribution mismatches and outliers than direct Lloyd–Max designs. Accordingly, we employ a power-law-based companding strategy similar to [26, 27], in which the distribution is first nonlinearly transformed as

$$P_{\mathbf{T}}(T) \mapsto \frac{P_{\mathbf{T}}(T)^\gamma}{\int P_{\mathbf{T}}(u)^\gamma du}, \quad 0 < \gamma < 1.$$

With $\gamma \in (0, 1)$, this transformation flattens concentrated regions of the distribution and expands underrepresented regions, thereby facilitating more balanced quantization. Uniform quantization is then performed in the transformed domain, and the outputs are mapped back to yield the quantized sequence $\{\hat{T}_n\}$. These quantized encodings are subsequently used in the reconstruction stage to estimate $f(t)$.

223 4. Experimental Results

224 In this section, we validate our theoretical findings for synthetically gen-
 225 erated signals and for signals acquired from practical applications (cf. the
 226 supplementary material).

227 For simulations, we generated synthetic signals as sums of sinc functions
 228 with spectral support up to 50 Hz and amplitude range $[-1, 1]$, restricted to
 229 the interval $[-0.45s, 0.45s]$. For the TEM, we set $b = 1.2$ and $\Delta = 0.0015$ to
 230 preserve the guarantees of perfect reconstruction.

231 For comparison, we also implemented a non-uniform sampling (NUS)
 232 scheme, where samples $\{f(t_n), t_n\}$ were taken at the IF-TEM firing instants,
 233 ensuring adaptivity similar to TEM. In both IF-TEM and NUS, the time
 234 encodings were quantized using a power-law-based compander. Across all
 235 experiments, we found that setting $\gamma = 0.1$ yielded the lowest error. In NUS,
 236 the amplitudes $f(t_n)$ were quantized using an NUQ designed via the Lloyd-
 237 Max algorithm, which consistently yielded the best results in this setting.
 238 For the NUQ, we analyzed the amplitude statistics of the generated synthetic
 239 signals by averaging them, then used the transformation as in Theorem 1.

240 We also compare the performance of uniform sampling (US), using sinc
 241 interpolation, with uniform and non-uniform quantization. The US is per-
 242 formed and reconstructed as $f(t) = \sum_{n=-\infty}^{\infty} f(nT_s) \text{sinc}\left(\frac{t-nT_s}{T_s}\right)$, where
 243 $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, and $T_s \leq \frac{\pi}{\Omega_0}$.

244 Fig. 3 presents the primary comparison among all six schemes: IF-TEM
 245 and NUS with uniform and non-uniform quantization, as well as US with

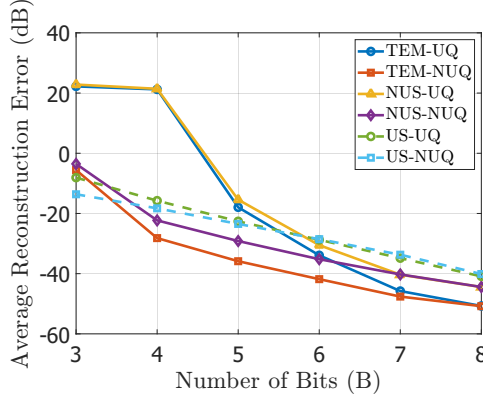
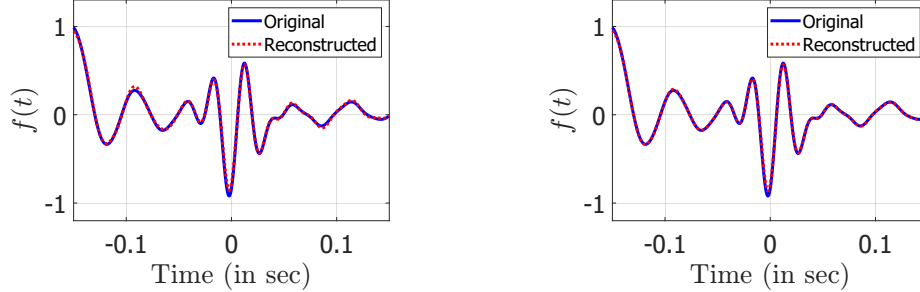


Figure 3: NMSE comparison of TEM and NUS under uniform and non-uniform quantization.

amplitude UQ and NUQ, for the same class of signals and the same number of samples. For NUS, the notation B bits indicates that B bits are used for time encodings and an additional B for amplitudes, giving a total of $2B$ bits—twice the communication cost of TEM. The results show that NUQ consistently outperforms UQ in both IF-TEM and NUS. Moreover, TEM-NUQ achieves lower error than NUS-NUQ, despite using half the number of bits. In comparison, for the US scheme, we observe a negligible difference between UQ and NUQ. Overall, TEM-NUQ performs significantly better than all competing schemes, including US with both UQ and NUQ.

Finally, Fig. 4 visualizes reconstruction quality. With -30 dB NMSE as the threshold for acceptable reconstruction, this level of accuracy is achieved with 4 bits in TEM-NUQ, 6 bits in TEM-UQ, 10 bits in NUS-NUQ, and 12 bits in NUS-UQ. This highlights the superior rate-distortion tradeoff offered by the proposed TEM-NUQ framework. Further experimental results on sig-



(a): Non-Uniform quantization in NUS with 4 bits (NMSE = -23.95 dB). (b): Non-Uniform quantization in TEM with 4 bits (NMSE = -29.8 dB).

Figure 4: Reconstruction error comparison for IF-TEM and NUS under NUQ.

nals captured on real-world data are reported in the supplementary material. In both real and synthetic data experiments, we observe that TEM-NUQ achieves lower reconstruction error at lower bit rates, demonstrating both accuracy and efficiency. Overall, these results highlight the benefits of incorporating statistical knowledge into quantizer design and position TEMs as a compelling alternative to conventional sampling and quantization schemes.

Having shown theoretically and through simulations that NUQ in TEM is an appealing framework, we would like to take our insights one step further by modifying the TEM to achieve a concentration in firings and capitalize on this to similarly design an NUQ. The following sections build upon this problem.

5. Design and Analysis of the LB-IF-TEM

In this section, we introduce the proposed LB-IF-TEM and show that, by enabling bias to closely track the signal, the firing interval's variance can be minimized, enabling us to design an efficient NUQ. First, we discuss the

275 selection of the bias, keeping in mind the guarantees of perfect reconstruction,
 276 and then derive bounds on the oversampling.

277 5.1. Linear bias proposition

278 In this scheme, the bias is a piecewise linear function that maintains the
 279 same slope but shifts after each firing. The bias for $t \in [t_n, t_{n+1}]$ is denoted
 280 as $b_n(t)$. Bias is called feasible if $f(t) + b_n(t) > 0$ for $t \in [t_n, t_{n+1}]$. To derive
 281 the linear bias, we consider the bias calculation after the firing instant t_n .
 282 We assume that within the previous firing interval, $t \in [t_{n-1}, t_n]$, the bias was
 283 feasible and linear. Next, we discuss the main principle of bias calculation.

284 Our approach to bias calculation is based on the following characteristics
 285 of bandlimited signals. If $f(t_n)$ is known, then by using the fact that $|f'(t)| \leq$
 286 ϵ we have $-\epsilon(t - t_n) + f(t_n) \leq f(t) \leq \epsilon(t - t_n) + f(t_n)$, for $t \geq t_n$. This implies
 287 that the signal lies between two straight lines with slopes $\pm\epsilon$. Importantly,
 288 the bias will increase along the straight line with slope ϵ . Since the biased
 289 signal should be positive, if we choose the bias as $b_n(t) = \epsilon(t - t_n) - f(t_n) + \mu$,
 290 where $\mu > 0$, then we have that $f(t) + b_n(t) > 0$ for $t \geq t_n$.

291 The approach mentioned previously requires the knowledge of $f(t_n)$. How-
 292 ever, an IF-TEM approach measures the signal's averages rather than the
 293 instantaneous values. To address the issue, we first demonstrate that the
 294 minimum and maximum values of $f(t_n)$ can be determined from the previ-
 295 ous TEM measurement. Then, by using the minimum value of $f(t_n)$, we
 296 update the bias rule as discussed next.

297 From the previous time encodings $\{t_n, t_{n-1}\}$ and bias $b_{n-1}(t)$, the average
 298 value of the signal $\hat{f}_{n-1}$ is determined using (4) as follows:

$$\hat{f}_{n-1} = \frac{1}{T_{n-1}} \int_{t_{n-1}}^{t_n} f(t) dt = \frac{\Delta}{T_{n-1}} - \frac{\int_{t_{n-1}}^{t_n} b_{n-1}(t) dt}{T_{n-1}}. \quad (10)$$

299 Since $f(t)$ is continuous (being band-limited with finite energy), by the Mean
 300 Value Theorem, there exists a point $\xi_n \in (t_{n-1}, t_n)$ such that $f(\xi_n) = \hat{f}_{n-1}$.
 301 From point ξ_n , the signal can increase or decrease with a maximum slope of
 302 $\pm\epsilon$. By drawing lines with these slopes starting at $\hat{f}_{n-1}$ at ξ_n , and intersecting
 303 them with the vertical at t_n , we determine a range for $f(t_n)$ (cf. Fig. 5).
 304 Specifically, we obtain the bound $f(\xi_n) - (t_n - \xi_n)\epsilon \leq f(t_n) \leq f(\xi_n) + (t_n - \xi_n)\epsilon$.

305 Now, instead of using the exact value of $f(t_n)$ for bias calculation, if we
 306 consider its minimum value, as derived in our previous work [20], the bias
 307 would still be feasible. Hence, the proposed bias after the firing at $t = t_n$ is

$$b_n(t) = \epsilon(t - t_n) - \hat{f}_{n-1} + \epsilon \frac{(t_n - t_{n-1})}{2} + \mu. \quad (11)$$

308 The proposed linear bias is parametrized by the threshold Δ and the bias shift
 309 factor μ , as seen from equations (10) and (11). In the following, we discuss
 310 how the change in the parameters affects the firing rates and compare them
 311 with existing results.

312 5.2. Interval Bounds and Parameter Selection

313 In this subsection, we characterize the effect of the linear bias on the
 314 firing intervals and provide bounds on the oversampling factor. Specifically,
 315 we demonstrate that the upper and lower firing bounds can be user-defined,
 316 allowing one to choose the parameters accordingly. Our main results are
 317 summarized in the following theorem.

318 **Theorem 2** (Bounds on Firing Intervals Under Linear Bias). *Consider sig-*
 319 *nals in the class $\mathcal{B}_{\Omega_0, c}$ sampled using the LB-IF-TEM with a linear time-*
 320 *varying bias of the form in (11). Suppose a user prescribes bounds $0 < \alpha <$*
 321 *$\beta \leq 1$ on the firing intervals such that*

$$\alpha T_{Nyq} \leq T_n \leq \beta T_{Nyq} \quad \forall n. \quad (12)$$

322 *Then, the LB-IF-TEM parameters Δ and μ must be chosen as*

$$\Delta = \frac{\epsilon \alpha \beta T_{Nyq}^2 (\beta + \alpha)}{\beta - \alpha}, \quad \text{and} \quad \mu = \frac{\epsilon \alpha T_{Nyq} (\beta + \alpha)}{\beta - \alpha}, \quad (13)$$

323 *in order to guarantee the bounds in (12).*

324 *Proof.* We prove the interval bounds in two parts by finding $T_{\max}$ and $T_{\min}$
 325 such that $T_{\min} \leq T_n \leq T_{\max}$. For perfect reconstruction, we should have
 326 $T_{\max} = \beta T_{Nyq}$, where $0 < \beta \leq 1$, this condition arises from the use of the
 327 Bernstein inequality in the proof of the reconstruction algorithm, the details
 328 of which are presented in the supplementary material.

340 and resulting in faster firing. Hence, we consider the areas depicted in purple
 341 and green in Fig. 5, and equate them to the threshold Δ as

$$\Delta = \frac{1}{2}T_{\min} (\epsilon T_{n-1} + \epsilon T_{n-1} + 2T_{\min}\epsilon) + \mu T_{\min}. \quad (15)$$

342 Solving the quadratic equation in (15) we have

$$T_{\min} = \frac{-(\epsilon T_{n-1} + \mu) + \sqrt{(\epsilon T_{n-1} + \mu)^2 + 4\epsilon\Delta}}{2\epsilon}. \quad (16)$$

343 Note that the right-hand side of (16) is a monotone decreasing function of the
 344 previous interval T_{n-1} , obtained by analyzing the differentiation with respect
 345 to T_{n-1} . Notice that $\frac{dT_{\min}}{dT_{n-1}} = \frac{-1}{2} + \frac{\epsilon T_{n-1} + \mu}{2\sqrt{(\epsilon T_{n-1} + \mu)^2 + 4\epsilon\Delta}} \leq 0$. Hence the smallest
 346 possible $T_{\min}$ occurs when the previous interval is maximal: $T_{n-1} = \beta T_{\text{Nyq}}$.
 347 We now enforce the condition $t_{\min} = \alpha T_{\text{Nyq}}$. Substituting these into (16) and
 348 rearranging yields the relation $\Delta = \alpha\epsilon T_{\text{Nyq}}^2(\beta + \alpha) + \alpha T_{\text{Nyq}}\mu$. Using (14) to
 349 eliminate μ we obtain $\Delta = \alpha\epsilon T_{\text{Nyq}}^2(\beta + \alpha) + \frac{\alpha}{\beta}\Delta$. Solving for Δ yields the
 350 closed-form

$$\Delta = \frac{\alpha\beta\epsilon T_{\text{Nyq}}^2(\beta + \alpha)}{\beta - \alpha} \quad (17)$$

351 and therefore, using (14),

$$\mu = \frac{\Delta}{\beta T_{\text{Nyq}}} = \frac{\alpha\epsilon T_{\text{Nyq}}(\beta + \alpha)}{\beta - \alpha}. \quad (18)$$

352 With Δ and μ chosen according to (17)–(18), the worst-case minimal
 353 interval equals $T_{\min} = \alpha T_{\text{Nyq}}$ and the maximal interval equals $T_{\max} = \beta T_{\text{Nyq}}$;
 354 which completes the proof. $\square$

355 The result shows that α and β can be freely selected (subject to $\alpha < \beta \leq$
 356 1), enabling explicit control of the oversampling ratio. Unlike the VB-IF-
 357 TEM, the LB-IF-TEM imposes no upper limit on achievable oversampling.

358 Moreover, event-driven sampling excels for bursty signals but over-samples
 359 bandlimited signals in high-amplitude regions. By tracking the signal with
 360 a linear bias, the integrator sees an almost constant input, yielding near-
 361 uniform, amplitude-insensitive firings. This design offers explicit control of
 362 sensitivity (via Δ, μ) and removes VB-IF-TEM’s oversampling limits—better
 363 matching the uniformly distributed information in bandlimited signals and
 364 enabling more efficient low-bit NUQ.

365 In terms of reconstruction, $\{t_n\}$ are required, as well as $\hat{f}_n$, calculated
 366 from (10) [11]. Note that there is a slight increase in complexity, since we
 367 must also keep track of the integral of the bias. This is a trade-off between
 368 the complexity of the reconstruction algorithm and the information required
 369 for reconstruction. However, we would like to point out that, since our bias is
 370 a linear function, calculating its integral imposes a negligible computational
 371 burden when compared to the iterative steps required for recovery.

372 Our motivation in designing the LB-IF-TEM was to achieve more con-
 373 centrated firing times that enable efficient deployment of an NUQ scheme,
 374 and also address the limited oversampling capability of the VB-IF-TEM, as

Table 1: Comparison of $T_{\min}$, $T_{\max}$, and T_{range} across TEMs

TEM	$T_{\max}$	$T_{\min}$	T_{range}
CIF	$\frac{\Delta_c}{b-c}$	$\frac{\Delta_c}{b+c}$	$\frac{2c\Delta_c}{b^2-c^2}$
VB	$\sqrt{\frac{2\Delta_v}{\epsilon}}$	$(\sqrt{5}-2)\sqrt{\frac{2\Delta_v}{\epsilon}}$	$(3-\sqrt{5})\sqrt{\frac{2\Delta_v}{\epsilon}}$
LB	$\frac{\Delta_l}{\mu}$	$\frac{-(\epsilon\Delta_l+\mu^2)+\sqrt{(\epsilon\Delta_l+\mu^2)^2+4\epsilon\Delta_l\mu^2}}{2\epsilon\mu}$	$\frac{\Delta_l}{\mu} - \frac{-(\epsilon\Delta_l+\mu^2)+\sqrt{(\epsilon\Delta_l+\mu^2)^2+4\epsilon\Delta_l\mu^2}}{2\epsilon\mu}$

375 demonstrated in Theorem 2. To this end, we compare the proposed LB-
 376 IF-TEM with the conventional IF-TEM (labelled as CIF), with constant
 377 bias [11] and the VB-IF-TEM [20]. Table 1 summarizes the parameter
 378 expressions for all three time-encoding mechanisms, along with the corre-
 379 sponding minimum and maximum firing intervals and the interval range
 380 $T_{\text{range}} = T_{\max} - T_{\min}$. Here, Δ_c and b denote the threshold and bias for the
 381 conventional IF-TEM, respectively, while Δ_v and Δ_l represent the thresholds
 382 in the VB-IF-TEM and LB-IF-TEM, and μ is the bias shift factor in the LB-
 383 IF-TEM. For a more direct comparison, the table also lists specific parameter
 384 values of $\Delta_c = 0.02$, $b = 1.5$, $\Delta_v = 0.0157$, $\Delta_l = 0.0314$, $\mu = 3.14$ used in our
 385 simulations. The parameters are selected such that all three schemes achieve
 386 a normalized mean-squared error (NMSE) of -50 , dB in reconstructing ban-
 387 dlimited signals, while maintaining $T_{\max} = T_{\text{Nyq}} = 10$ msec. In this case,
 388 T_{range} s (in msec) for CIF, VB, and LB are 8.8, 7.6, and 5.9, respectively.

389 From Table 1, it is evident that the LB-IF-TEM produces a narrower
 390 range of firing intervals compared to the other methods. To illustrate the fir-
 391 ing behavior with signal variation, Fig. 6 shows a bandlimited signal (blue),
 392 its reconstruction using the LB-IF-TEM (red), and the corresponding firing

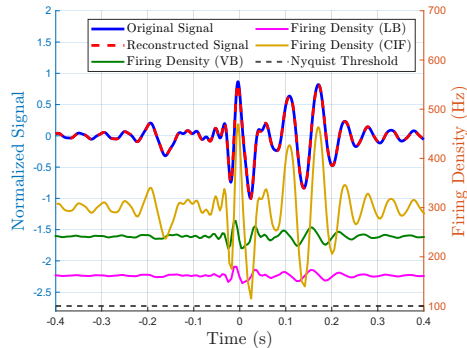


Figure 6: Firing rates for the different TEM variants. Observe the lower variation and low rate for the LB-IF-TEM and VB-IF-TEM. It is this low variation in firing that the NUQ capitalizes on, and the low firing rate that results in increased information throughput.

densities (computed as $1/T_n$). The densities in the LB-IF-TEM are concentrated within approximately 10 Hz, whereas the spreads are about 20 Hz and 50 Hz for the VB-IF-TEM and conventional schemes, respectively.

While limiting the variation in firing intervals can be disadvantageous for spiky or highly localized signals that benefit from adaptive timing, it is advantageous for smoothly varying bandlimited signals, where information is more uniformly distributed in time. This validates our design intuition of concentrating firings to improve quantization efficiency. In the next section, we analyze the simulation results to quantify the resulting gain in communication cost.

6. NUQ and UQ on variants of TEM

We evaluate the proposed LB-IF-TEM using bandlimited signals with bandwidth $\Omega_0 = 100\pi$ and bound on amplitude as $c = 1$, defined over the interval $[-0.45, 0.45]$ seconds. Under Shannon–Nyquist sampling, perfect re-

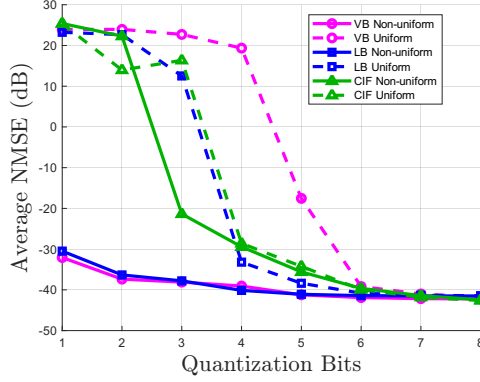


Figure 7: Comparison of NMSE versus bit budget for different IF-TEM schemes under UQ/NUQ.

407 construction would require $2 \times (\Omega_0/2\pi) \times (\text{time interval}) = 90$ samples. To
 408 study quantization performance, the parameters of the conventional, VB-,
 409 and LB-IF-TEMs are tuned to achieve an NMSE of -50 dB. The corre-
 410 sponding numbers of firings required are 266, 214, and 144, respectively.

411 To design the NUQ, we first construct histograms of the firing intervals
 412 (over 100 randomly generated signals) and generate codebooks using both
 413 the Max-Lloyd algorithm [28] and a power-law-based companding strategy
 414 [29]. For each TEM framework, the NUQ configuration yielding the lower
 415 average NMSE (over 100 signals) is retained.

416 Fig. 7 compares the NMSEs for UQ and NUQ as a function of bit rate.
 417 The NUQ consistently achieves lower error than the UQ in both VB-IF-
 418 TEM and LB-IF-TEM, particularly in the low-bit regime. In contrast, the
 419 performance gain is less pronounced for the conventional IF-TEM (labelled as
 420 CIF), as its constant bias leads to greater variance in firing intervals. Both
 421 VB-IF-TEM and LB-IF-TEM achieve comparable reconstruction accuracy

422 under NUQ; however, the LB-IF-TEM requires roughly $1.5\times$ fewer firings to
423 reach the same NMSE.

424 We also observe that at 1-bit quantization, both LB-IF-TEM and VB-IF-
425 TEM achieve errors of the order of -30 dB NMSE, corresponding to good
426 reconstruction. This further shows that there is a peak in firings, which the
427 NUQ capitalizes on.

428 These results confirm that the proposed LB-IF-TEM achieves lower dis-
429 tortion and reduced communication cost by effectively localizing the firing
430 intervals and exploiting non-uniform quantization more efficiently.

431 7. Conclusion

432 We presented a framework for non-uniform quantization of IF-TEM encod-
433 ings by exploiting the statistical relation between signal amplitudes and firing
434 intervals, deriving the induced interval distribution and designing a power-
435 law-based NUQ that outperforms uniform quantization and, when compared
436 with NUS, achieves lower reconstruction error at roughly half the bit rate.
437 We further introduced the linear-bias IF-TEM (LB-IF-TEM), established
438 bounds on its oversampling range, and showed that a bias tracking the sig-
439 nal produces low-variance firing intervals well suited to NUQ, with experi-
440 ments demonstrating superior rate-distortion performance and reduced com-
441 munication cost—even with 1-bit quantization at modest oversampling—thus
442 providing a low-power, communication-efficient framework for time-based en-
443 coding of both bursty and smoothly varying bandlimited signals.

444 References

- 445 [1] A. Gersho, R. M. Gray, Vector Quantization and Signal Compression,
446 Springer, 2012.
- 447 [2] M. Ben-Romdhane, C. Rebai, A. Ghazel, P. Desgreys, P. Loumeau, Low
448 power data conversion based on non-uniform sampling for multistandard
449 receiver, in: Proc. Int. Conf. Design & Tech. Integrated Sys. Nanoscal
450 Era, 2009, pp. 261–265.
- 451 [3] F. Marvasti, Nonuniform Sampling: Theory and Practice, Springer,
452 2001.
- 453 [4] R. Kumaresan, N. Panchal, Encoding bandpass signals using zero/level
454 crossings: A model-based approach, IEEE Trans. Audio Speech Lang.
455 Process. 18 (1) (2009) 17–33.
- 456 [5] D. Gontier, M. Vetterli, Sampling based on timing: Time encoding ma-
457 chines on shift-invariant subspaces, Appl. Comput. Har. Anal. 36 (1)
458 (2014) 63–78.
- 459 [6] H. Naaman, S. Mulleti, Y. C. Eldar, FRI-TEM: Time encoding sampling
460 of finite-rate-of-innovation signals, IEEE Tran. Signal Process. 70 (2022)
461 2267–2279.
- 462 [7] A. Omar, A. Cohen, Adaptive integrate-and-fire time encoding machine,
463 in: Proc. European Signal Process. Conf. (EUSIPCO), IEEE, 2024, pp.
464 2442–2446.

- 465 [8] H. Naaman, S. Mulleti, Y. C. Eldar, A. Cohen, Time-based quantiza-
466 tion for FRI and bandlimited signals, in: European Signal Processing
467 Conference (EUSIPCO), 2022, pp. 2241–2245.
- 468 [9] H. Naaman, N. I. Bernardo, A. Cohen, Y. C. Eldar, Time encoding
469 quantization of bandlimited and finite-rate-of-innovation signals, *IEEE*
470 *Acc.* (2025).
- 471 [10] A. A. Lazar, Time encoding with an integrate-and-fire neuron with a
472 refractory period, *Neurocomputing* 58 (2004) 53–58.
- 473 [11] A. A. Lazar, L. T. Toth, Perfect recovery and sensitivity analysis of
474 time encoded bandlimited signals, *IEEE Trans. Circ. Sys. I: Reg. Papers*
475 51 (10) (2004) 2060–2073.
- 476 [12] N. T. Thao, D. Rzepka, Time encoding of bandlimited signals: Re-
477 construction by pseudo-inversion and time-varying multiplierless FIR
478 filtering, *IEEE Trans. Signal Process.* 69 (2020) 341–356.
- 479 [13] N. Fu, H. Zhang, S. Yun, Z. Wei, L. Qiao, Time-based finite-rate-of-
480 innovation sampling for variable-pulse-width signal, *IEEE Transa. Inst.*
481 *Meas.* 73 (2024) 1–9.
- 482 [14] D. Florescu, A generalized approach for recovering time encoded signals
483 with finite rate of innovation, *IEEE Trans. Signal Process.* 73 (2025)
484 1862–1876.

- 485 [15] S. Tarnopolsky, H. Naaman, Y. C. Eldar, A. Cohen, Compressed IF-
486 TEM: Time encoding analog-to-digital compression, arXiv preprint
487 arXiv:2210.17544 (2022).
- 488 [16] S. Mulleti, T. Zirtiloglu, A. Tan, R. T. Yazicigil, Y. C. Eldar, Power-
489 efficient sampling: Towards low-power analog-to-digital converters,
490 IEEE Signal Processing Magazine 42 (2) (2025) 106–125.
- 491 [17] A. Arora, K. Yashaswini, S. Mulleti, Linear-bias time encoding for low-
492 rate quantized representation of bandlimited signals, arXiv preprint
493 arXiv:2511.09007 (2025).
- 494 [18] K. Yashaswini, A. Arora, S. Mulleti, A non-uniform quantization
495 framework for time-encoding machines, arXiv preprint arXiv:2511.02728
496 (2025).
- 497 [19] D. Florescu, Model-driven quantization for time encoding machines, in:
498 Proc. Int. Conf. Sampling Theory Appl. (SampTA), 2023, pp. 1–5.
- 499 [20] A. Arora, S. Mulleti, A lowrate variable-bias integrate-and-fire time
500 encoding machine, in: Proc. Int. Conf. Acous. Speech Signal Process.
501 (ICASSP), 2025, pp. 1–5.
- 502 [21] A. J. Kamath, C. S. Seelamantula, Differentiate-and-fire time-encoding
503 of finite-rate-of-innovation signals, in: Proc. Int. Conf. Acous. Speech
504 Signal Process. (ICASSP), IEEE, 2022, pp. 5637–5641.

- 505 [22] A. Papoulis, Signal Analysis, McGraw-Hill, 1977.
- 506 [23] H. G. Feichtinger, K. Gröchenig, Theory and practice of irregular sam-
 507 pling, in: Wavelets: Mathematics and Applications, CRC Press, Boca
 508 Raton, FL, USA, 1994, pp. 305–363.
- 509 [24] T. Beyrouthy, L. Fesquet, R. Rolland, Data sampling and processing:
 510 Uniform vs. non-uniform schemes, in: Proc. Int. Conf. Event-based Con-
 511 trol Comm. Signal Process. (EBCCSP), 2015, pp. 1–6.
- 512 [25] A. Papoulis, S. U. Pillai, Probability, Random Variables, and Stochastic
 513 Processes, 4th Edition, McGraw-Hill, 2002.
- 514 [26] J.-F. Bercher, Source coding with escort distributions and rényi entropy
 515 bounds, Physics Letters A 373 (36) (2009) 3235–3238.
- 516 [27] C. Tsallis, A. R. Plastino, R. F. Alvarez-Estrada, Escort mean values
 517 and the characterization of power-law-decaying probability densities,
 518 Journal of Mathematical Physics 50 (4) (2009).
- 519 [28] S. Lloyd, Least squares quantization in pcm, IEEE Transactions on In-
 520 formation Theory 28 (2) (1982) 129–137.
- 521 [29] B. Smith, Instantaneous companding of quantized signals, Bell System
 522 Technical Journal 36 (3) (1957) 653–709.